VU Medical Zone

Z00-503 All past papers
Created By Amaan Khan

Objective

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1) The investigation of thermal maturation is now a routine **petroleum exploration technique**
2) The words science come From **Latin** scientia, meaning "knowledge"
3) Turbidity and light penetration are **INVERSLY** proportional to each other.
4) The fossil record is **rarely** complete
5) The Paleontology Word Originates from **Greek** "palaios-old", "ontos-creature" "logos-study"
6) The Earth today can be divided into **six** main provinces.
7) Evolution takes place both within **Species lineages** at the time of **speciation** =(both)
8) Transformation of rocks leads to **destroying the fossil**
9) Identification of fossils species is done according to their, **none of the given options**.
10) Humans are more closely related to **chimpanzees**.
11) The principle of **cross-cutting relationships:**
"If a body or discontinuity cuts across astratum, it must have formed after that stratum".
12) **Aristotle** said that **"All** *men are mortal. Socrates is a man. Therefore Socrates is mortal."*
13) Precise biotic and morphological changes along phylogenetic lineages. **The Work of John Phillips.**
14) Skeletons that are made from several parts may become **disarticulated** separated into components parts
15) Paleontologists should be **careful** when they attempt to reconstruct a whole plant or animal, and try to understand its biomechanics, when they have just a few bones or bits of twigs
16) **Charles Lyell** was an antiprogressionist
17) **Charles Darwin** "imperfection of the geological record"
18) **"Noah's arks".** Example of The transit of India from Gondwana to Asia
19) Important factor in driving evolutionary change **Climate**.
20) Each population, or set of populations, would start out with a **different gene pool**.
21) Right order of The -Conenoses is **biocoenosis, > thanetocoenosisv>Taphocoenosis**.
22) **Cuvier** is father of comparative anatomy.
23) Best amount of fossils that are present at sediment water interface. **both (Infaunal organisms Epifaunal organisms)**
24) Soil rich in calcium carbonate) and evaporates **dry, arid** climate
25) A succession of contiguous formations with common characteristics is called a **group**
26) Incomplete remains of **Neandertal man** in 1856.
27) Condition that are necessary for survival of hard part in the sediments. **Chemical condition**.
28) Fossils Form can tell us about (**All of these**).
29) Paleontology lies on the border between **Biology and Geology**.
30) Wing of bird related to term **ANALOGY**.
31) The remains of the activity of ancient organisms, such as burrows and tracks. **Trace fossils**.
32) Shelled organism lived in fresh water have **high calcification** than marine water.

33) Human baby tail and excessive hairs related to term **atvasims**.
34) Population growth and resources availability are **inversely** to each other
35) Life on earth is **4 billion years**.
36) The Spirallians are morphologically diversed clade of protostome animals including all of the following except……..**Cnidarians**
37) life was mainly affected by …..…..**predation**
38) Amphibians and reptiles evolved in …..**Triassic**
39) Lucy fossil old myb…. **3.2 million years**
40) Reptilian ancestors of mammals had… **synapsid skull**
41) The most advance mollusks are……..**Cephalopods**
42) Amniotes radiated in……..**Late carboniferous**
43) In early corocodilian were most diverse……..**Jurassic and Cretaceous**
44) Crocodile radiate in…..**jurassic**
45) First fossil evidence of bipedalism…… **Australopithecus afarensis**
46) Crocodile more diverse… **Jurassic and cretaceous**
47) Oldest known bird found in…….**Upper Jurassic sediments**
48) The oldest human was found is approximately lived _____ago…….**6Mya**
49) German paleontologist ---------- established a classification of trace fossil based on behavior ……..**Adolf Seilacher**
50) ……..are the observation of the appearance and disappearance of species……..**Patterns**
51) Which Scientist translated Cuvier's idea into English in 1813……**Robert Jameson**
52) ……..argues that all areas connected by one of these tracks had originally formed a single, continuous are that was inhabited by group concerned…….**Panbiogeography**
53) Briggs used pathogen of endamicity of coastal faunas to identify location where there appears to be a zone of unusually rapid faunal change and then used this to distinguish ---------- **zoogeographical regions**
54) Brigg distinguish _____ zoogeographical regions…….**23**
55) The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resources is called……..**Fundamental Niche**
56) ____noted that the range boundaries of Amazonia birds seemed to coincide with the patterns of river……**Alfred Wallace**
57) Geological study of New Caledonia show that the Island was submerged for long period in the Paleocene……..Eocene so that Island was not available for colonization until about……**37 mya**
58) The ancestors of mammoths and modern Asian elephant probably evolved in……**Africa**
59) During the early Devonian 380mya the fossil record suggest that all the early amphibians and reptile groups evolved in the continents called……..**Euramerica**
60) All of the following part of Gondwana Except……..**North America**
61) Many studies have demonstrated that the richness of the species particularly plant increases with ---**Altitude**
62) Evolution a process by which new species are generated it essentially involve……**Genetic Variation and Subsequent selection**
63) The trade winds found in the both northern and southern hemisphere meet in the region of equators this is known as……..**Intertropical Convergence zone (ITCZ)**
64) _____ is a term sometimes used to describe waves of biological invasions that has occurred in the wake of human invasions by either accidental, transport or deliberate introduction ……**Ecological Imperialism**
65) When two species are attempting to tap a resource in the same way and when that resource in short supply the two are said to be in……**Competition**
66) The theory of Island biogeography published in……..**1967**
67) All of the following are included in ecological biogeography except……**Altitude**

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68) In terms of diversification of life during the early Cambrian _____ was an important part of an ecosystem……**Predation**

69) The focus of human evolution was in…….**Africa**

70) The reptile ancestors of the mammals are……..**Synapsid**

71) Chondricthyans is fish……**Cartilaginous**

72) _____is a term sometimes used to describe the wave of biological invasion that has occurred……..**Ecological imperialism**

73) Biogeography is an integrative field of inquiry that unites concept and information from solving branches of science except……..Any option except these (**Ecology, Evolutionary Biology, Geology, Physical Geography**).

74) Biology of ocean……**marine biology**

75) _____in on origin of species where devoted to geographical distributions…….**Two Chapters**

76) The oldest substantial skeleton of praeanthropus afarensis lived approximately ---------- ago…….3**.2 Mya**

77) Early vertebrate skeletion…… **cartilage**

78) Certain species play key roles in the ecosystem, and when they are lost there is a sudden drop in the capacity of the ecosystem to function…..**Rivert hypthosis**

79) Paleontology Originate from Greek word "Ontos" which means……**Creature**

80) Today Extenction**….. 6th**

81) Archaeopetyx is true missing link between…….**Dinosaurs and Birds**

82) Trilobites dominated assemblages in England and Wales area were described by…….**William Smith**

83) Fossils in Fold belts are, except…….**In Excellent Shape**

84) Where an organism live matters in which of the filters is preservation of the complete fossil record……**Ecological Filters**

85) In Climate change in life history of earth has oscillated at least ---------- between icehouse and greenhouse conditions……..**5 times**

86) *Scala Naturae* is ----------**The Ladder of life**

87) When local ecological conditions affect the form of an organism during its life time and consequently fossil is called …….**Ecophenotypic Effects**

88) Rohde and Muller (2005) argued that there is____ periodicity in mass extinctions …….**62 myr**

89) Modification of the Oparin_____Haldane biochemical model is……**Hydrothermal Model**

90) ___Biomarker indicates that eukaryotes were present around at least 2.7 Ga **Lipid**

91) Chromistans Includes all of the following, except……..**Protozoa**

92) The Parazoan body plan is seen in…….**Sponges**

93) Fossils Lagerstattan are assemblages of…….**inadequate Preserved soft bodied**

94) The Spirallians are morphologically diversed clade of protostome animals including all of the following except……**Cinideria**

95) The most advanced mollusks are ……..**Cephalopods**

96) The "Belly footed" Mollusks are……**Gastropods**

97) The acme of the group Gastropoda with the radiation of the siphonal carnivorous neogastropods was in …….**Cenozoic**

98) Chelicerates are diversed and heterogeneous which includes….**All of the above (Mites, Scorpions, Spiders)**

99) Trilobites dominated assemblages in England and Wales area were described by **William Smith**

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Subjective

1) **Biogeography definition / Meant of Biography?**

Biogeography is the study of the distribution of species and ecosystems in geographic space and through geological time.

2) **Trace Fossi name types?**

Mold fossils (a fossilized impression made in the substrate - a negative image of the organism) Cast fossils (formed when a mold is filled in)
Trace fossils = ichnofossils (fossilized nests, gastroliths, burrows, footprints, etc.)

3) **Work of James Dana and Edward Fob on Ocean biogeography?**

American scientist **James Dana**, who later became an eminent geologist. His brief paper, published in 1853, divided the surface waters of the globe into several different zones based on their mean minimum temperature.
Three years later 1856, the British zoologist **Edward Forbes** published the first comprehensive work, recognizing five depth zones and 25 faunal provinces along the coasts of the continents.
He was the first to recognize the enormous Indo- Pacific faunal region.

4) **Explain redundant Species hypothesis?**

According to the *redundant species hypothesis*, the removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem. Here, the function of an ecosystem could refer to its overall productivity, its rate of nutrient cycling or its general self-sustainability.

5) **What are characteristic features of Pterosaurs? 3**

The pterosaurs were proficient flapping flyers, with a lightweight body, narrow hatchet-shaped skull and a long narrow wing supported on a spectacularly elongated fourth finger of the hand. Pterosaurs were covered with hair, and were almost certainly **endothermic**. Most pterosaurs fed on fishes caught in coastal seas, but others were insectivorous.

6) **Functions of Geographic Information System.**

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

7) **What are three types of radiated amniotes?**

Amniotes radiated during the Late Carboniferous, giving rise to three main clades. These are distinguished by the pattern of openings in the side of the skull, especially the **temporal openings** behind the eye socket. These three skull patterns diagnose the key clades among amniotes which are
i. The anapsid ("no arch") is the primitive state skull pattern, since there are no temporal openings. The Anapsida include various early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles. The two other skull patterns seen in amniotes are
ii. The synapsid ("same arch"), where there is a lower temporal opening. The Synapsida include the "mammal-like reptiles" and the mammals.
iii. The diapsid ("two arch") pattern, where there are two temporal openings. The Diapsida includes a number of early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and

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birds.

8) **Work of Leon Croizat in field of Biogeography.**

He amassed a vast array of distributional data, representing each biogeographical pattern as a line, or track, connecting its known areas of distribution. Croizat studied the distribution patterns of many unrelated taxa, and for each he drew lines or 'tracks' on the map linking the areas in which they are found. In many cases, these lines were similar enough in position to be combined as 'generalized tracks', shown here. He found that the tracks of many taxa, belonging to a wide variety of organisms, could be combined to form a generalized track that connected different regions of the world

9) **What are two key defining characters of Jawless fishes?**

Two key defining characters: The head and neural crest tissues. The neural crest appears in the early embryo as a strip of cells lying just below the outer skin, the ectoderm, of the embryo, above the line of where the backbone will develop.

10) **Oldest clear evidence for bipedalism?**

The oldest clear evidence for bipedalism was the find of some human tracks in volcanic ash from Tanzania, dated at 3.75 Ma. The oldest substantial skeletons, of *Praeanthropus afarensis*, come from rocks dated at about 3.2 Ma and also show clear anatomic evidence for advanced bipedalism, but still an ape-sized brain.

11) **What was the brain capacity of Homo habilis?**

The first species, *H. habilis,* lived in Africa from 2.4 to 1.5 Ma, and had a brain capacity of 630–700 cm3 in a body only 1.3 m tall. *H. habilis* may have used tools.

12) **Name the most famous fossils. 3**

One of the most famous fossils is *Archaeopteryx*, the oldest known bird. The first specimen was found in Upper Jurassic sediments in southern Germany in 1861, and was hailed as the ideal "missing link" or proof of evolution in action. Here was an animal with a beak, wings and feathers, so it was clearly a bird, but it still had a reptilian bony tail, claws on the hand, and teeth. Since 1861, nine more skeletons have come to light, the last two in 1992 and 2005.

13) **Name the old amniotes and where it was found?**

The oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada is known only from its skeleton; no amniotic eggs are known from the Carboniferous.

14) **Three amniotic layers in embryo and their functions.**

- Enclose the embryo **(the amnion)**
- Collect waste (**the allantois**)
- line the eggshell (**the chorion**)
- Yolk, a yellow material rich in protein

15) **Trace fossils excrement names, how they are different to other? 3**

***Fecal pellets and fecal strings*:** small pellets, usually less than 10 mm in length, or strings of excrement.
***Coprolites*:** discrete fecal masses, usually more than 10 mm in length, and usually the product of vertebrates.

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16) **The work of Albert Gunther, and Arbold Ortman in Biogeography 3**

In 1880, the British zoologist **Albert Gunther** published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes.

The German **Arnold Ortmann** published a similar work based on the distribution of crustaceans such as crabs and lobsters.

17) **Erythrosachus**

Small and large meat eaters such as *Erythrosuchus* appeared, one of the first of the archosaurs, a group that was later to include the dinosaurs, pterosaurs, crocodilians and birds.

18) **Traces on bedding planes**

• **Tracks**: sets of discrete footprints, usually formed by arthropods or vertebrates
• **Trails:** continuous traces, usually formed by the whole body, either trave

19) **Describe the Island biogeography of New Candolle 3**

Another early contributor was Candolle, who pointed out that the age, climate and degree of isolation of an island, and whether or not it was volcanic, would also affect the diversity of its flora. Nevertheless, the sheer variety and volume of the works on island biogeography published by Alfred Wallace mark him as the real founder of studies on this subject.

20) **island biogeography...2**

Island biogeography is the study of the species composition and species richness on islands

21) **nannoplankton..2**

Defined as plankton less than 63 μm across, the smallest standard mesh size for sieves • Includes organicwalled and siliceous forms

22) **exual dimorphism...3**

dimorphism, in which males and females may show different sizes, and different specialized features variations within species.

Growth stages, where there may be quite different larval and adult stages, or where body form alters during Variations within the Species.

23) **Ecophenotypic effects**

Where local ecological conditions affect the form of an organism **during its lifetime**

24) **Oldest fossil n their ages..5**

The oldest eukaryote fossil may be Grypania, • A coiled, spaghetti-like organism that has been reported from rocks as old as 1.85 Ga

The oldest locomotory trace fossils are from about 550 Ma northwest Russia • Fecal strings from rocks some 600 Ma suggesting Bacteria in rocks up to 3.2 Ga, indicated by stromatolites

25) **Deuterostomes 5 marks**

"**Second mouth**" however, have a mouth arising from a secondary opening • True blastopore **develops as anus** • Echinoderm– **Hemichordate Chordate (ppts)**

*Deuterostomia, (Greek: "second mouth"), group of animals—including those of the***phyla** **Echinodermata** *(e.g.,* **starfish**, **sea urchins**), **Chordata** *(e.g.,* **sea squirts**,**lancelets**, *and* **vertebrates**), **Chaetognatha** *(e.g., arrowworms), and* **Brachiopoda***(e.g., lamp*

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shells)—classified together on the basis of embryological development (google)

26) **What is paleontology(2)**

The branch of science concerned with fossil animals and plants. • the study of fossils to determine organisms' evolution & ecology

27) **How anatomic filters helps to preserve complete fossil record(3)**

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. Entirely soft-bodied organisms, such as worms and jellyfish, are only preserved in rare cases.

28) **What are the four characteristics of phylum Mollusca(5)**

Unsegmented soft body with bilateral symmetry.
Presence of an internal or external shell.
A toothed tongue (made mostly of chitin) called the radula. A mantle which is a fold in the body wall that lines the shell. Muscular foot (and/or tentacles in some).

29) **What are the big five mass extinction events(5)**

During mass extinctions, 20–90% of species were wiped out; these include a broad range of organisms, and the events appear to have happened rapidly. It is difficult to study mass extinctions in the Precambrian • There have been a Neoproterozoic event between the Ediacaran and Early Cambrian faunas. The **"**big five**"** Phanerozoic mass extinctions occurred in the

- End-**Ordovician**
- Late **Devonian**
- End-**Permian**
- End-**Triassic**
- End-**Cretaceous**

30) **who introduce the concept of induction and explain it 2**

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature. Induction

31) **Define speciation and write name of two models2**

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population.
"The process of splitting of a population to form two species is speciation **"**. •The allopatric • (**"**other homeland**"**) model
• The geographic model (based on the establishment of geographic barriers

32) **write the name of animal groups who hits by end permian mass extinction event.3**

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic eruptions, leading to acid rain and global anoxia.(ppts)
During the period a number of animal groups became extinct, including the trilobites, tabulate and rugose corals, and blastoids (see mass extinction). **Amphibians** *and* **reptiles** *continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as the dominant* **plants***. "Permian. (google)*

33) **What is micropaleontology? 2**

A multidisciplinary science, focused on the study of microorganisms or the microscopic parts of

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larger organisms

34) **What are the Trilobitomorpha? what are their different body parts?**

3 Highly derived arthropods

- lack specialized mouthparts
- Tagmata comprising the cephalon, thorax and pygidium
- Trilobitomorph appendages

- Common: throughout the Paleozoic
- Extinction: end of the Permian

35) **What is "Great Ordovician Biodiversification Event" (GOBE)? 5**

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE.

Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased

Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

36) **Echinoderm Water vascular System?**

The **water vascular system** is a hydraulic **system**used by **echinoderms**, such as sea stars and sea urchins, for locomotion, food and waste transportation, and respiration. The **system** is composed of canals connecting numerous tube feet.

37) **Any 2 Models about Origin of Life?**

Scientific models: Some of them now rejected by the evidence • Others still available as potentially valid

hypotheses: 1. Spontaneous generation 2. Inorganic model. 3. Extraterrestrial origins 4. Biochemical model

5. Hydrothermal model.

The inorganic model: Complex organic molecules arose gradually on a preexisting, nonorganic replication platform • Silicate crystals • Graham CairnsSmith of Glasgow University, 2007 • not conclusive

38) **Define Paleontology?**

Paleontology, [pā'lē-ŏn-tŏl'ə-jē] The scientific study of life in the geologic past, especially through the study of animal and plant fossils.

39) **Define Speciation. Two Forms of Speciation?**

Speciation often occurs by the establishment of a barrier, and the isolation of part of a previously interbreeding population. Evolution takes place both within • Species lineages (phyletic gradualism) • at the time of speciation (punctuated equilibrium).

40) **Define induction?**

Sir Francis Bacon • The enquirer might hope to see common patterns among the observations, and these common patterns would point to an explanation, or law of nature.

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41) **Why the polychates have incomplete fossil? While paddle worm have complete fossil record?** Polychaetes: A diverse polychaete fauna has been described from the Burgess Shale • Even contains Canada spinosa • Similar to some living polychaetes. Paddle worms: most complete fossil record • Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts.

42) **Define dimorphic organism with example?**

Dimorphic fungi are fungi that can exist in the form of both mold and yeast. Anexample is Penicillium marneffei, a human pathogen that grows as a mold at room temperature, and as a yeast at human body temperature.

43) **Anatomic filters?**

Organisms are likely to be preserved only if they have hard parts, a skeleton of some kind. • Entirely softbodied organisms, such as worms and jellyfish, are only preserved in rare cases.

44) **Island biography?**

Islands are areas of land that are not connected to a continent and are surrounded by water. Small islands are sometimes called cays, keys, or islets. A group ofislands is often called an archipelago. There are two main type of islands; continental islands and oceanic islands.

45) **What is biostratigraphy 2m**

Biostratigraphy is the establishment of fossil based successions and their use in stratigraphic correlation • Biozones: The main operational units of a biostratigraphy

46) **What is the known range of zone fossils 3m**

The known range of a zone fossil is the time between its first and last appearances in a specific rock section, •First appearance datum (**FAD**) • Last appearance datum (**LAD**)

47) **in polycheate, why the worms has a complete fossil record 3m**

Record is enhanced by the relatively common preservation of elements of the phosphatic jaw scolecodonts. *The earthworm, which has been around for* **600 million years**, *is spread across the earth and provides a basic biological model for many other species including humans came out top*

48) **What Is Micropaleontology?**

Micropaleontology is the branch of paleontology that studies microfossils, or fossils that require the use of a microscope to see the organism, its morphology and its characteristic

49) **What Are Helicoplacoids?**

Helicoplacus is the earliest well-studied fossil echinoderm. Fossil plates are known from several regions. Complete specimens were found in Lower Cambrian strata of the White Mountains of California. The animal was a cigar-shaped creature up to 7 centimetres long that stood upright on one end.

50) **Role Of Amateurs?**

Amateurs are not paid to work as paleontologists • Collaborations between different sciences is the key.

51) **Thermal Maturity?**

Helps in identifying the levels of thermal maturity of rocks. {**Thermal maturity** is the extent of

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heatdriven reactions that alter the composition of organic matter (e.g., conversion of sedimentary organic matter to petroleum or cracking of oil to gas.)}

52) Gaia's Hypothesis?

Gaia hypothesis: Earth as a living organism in equilibrium with its biosphere.

53) Formation of Bed and Member?

A bed, is a lithologically distinct layer within a member or formation and is the smallest recognizable stratigraphic unit. **A member** is a more local lithologic development, usually part of a formation.

54) Isthmus Of Panama?

Barrier for some organisms may provide a corridor for others. • The emergence of the Isthmus of Panama (3 Mya) • Connected North and South America • Separated the Atlantic and Pacific oceans.

55) Types Of Fossils?

Five types of fossils: (a) insect preserved in **amber**, (b) **petrified wood** (**permineralization**), (c) **cast** and**mold** of a clam shell, (d) pyritized ammonite, and (e) compression fossil of a fern. **(Google)** Body fossils: the partial or complete remains of plants or animals 2. Trace fossils: the remains of the activity of ancient organisms, such as burrows and tracks.

56) Diff scavenging and Decay?

scavenging	Decay
When large animals feed on dead plant or animal tissues, the process is termed **scavenging**	When microbes, such as fungi or bacteria, transform tissues of the dead organism, the process is termed **decay**

57) Diff Volatile and Refractories?

Volatile	Refractories
Soft parts of animals are made from volatiles, forms of carbon that break down readily	Refractories are the organic carbons much less liable to break down, such as cellulose.

58) Infaunal and Epifaunal?

Infaunal organisms live beneath the sediment–water interface

Epifaunal organisms live above it Thensediment–water interface

59) Allometry and Isometry?

Allometry is a well-known study, particularly in statistical shape analysis for its theoretical developments,
as well as in **biology** for practical applications to the differential growth rates of the parts of a living organism's body.
The word 'allometric' comes from the word for different measure in Greek. Most organisms grow allometrically. We then learned that **isometric** growth is growth in which, the proportions of the organism grow the same.This comes from the Greek word that means same measure. Very few organisms grow isometrically.

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60) Endosymbiotic Theory?

Margulis: 1970s. • Prokaryote consumed, or was invaded by, prokaryotes • Two species evolved together in a mutually beneficial way.

Endosymbiotic theory is an evolutionary **theory**that assumes **endosymbiosis** between prokaryotic**endosymbionts** and eukaryotic host cell is the**means** by which organelles such as mitochondria and chloroplasts occur within eukaryotic cells.

61) Insects and Corals

Insects	Corals
Insects first appeared during the Early Devonian and diversified rapidly • There are probably 10 million species of living insects. Insects had probably already evolved flight before the Mid Carboniferous, when giant dragonflies patrolled the forests.	**Corals** Most diverse and most threatened ecosystems • The coral reef • Shallow-water coral reefs in zone extending 30° north and south of equator • 30m or 18 °C.

62) Usefulness of Microfossils Record?

Studying **microfossils** has a lot of **uses**. **Microfossils** are **used to** determine how old a piece of rock is and determine if there is gas or oil in the area. They are also**used to** see what kinds of major geological events took place such as earthquakes or major weather changes such as ice storms

63) Origin Of Life In Early Precambrian World?

The Precambrian is divided into

• Hadean eon • Archaean eon • Proterozoic eon **Hadean Eon** spans from the origin of the Earth, 4.57 to about 4 Ga. Beginning of the Hadean • Temperatures on surface were too high • The crust was too unstable for any form of carbon based life to exist. As the Earth's surface cooled, the lithosphere, its rocky crust, began to differentiate as a cooler upper layer above the underlying asthenosphere. The oldest rocks are from Canada and are dated at 3.8–4 Ga, and some mineral grains from Australia have even been dated to 4.4 Ga Lasted from about 4 to 2.5 Ga. • The oldest sedimentary rocks from the Isua Group in Greenland, dated at 3.8–3.7 Ga. The **Proterozoic Eon**, from 2.5 Ga to 542 Ma • 2.4 Ga, atm oxygen levels rose to onehundredth or onetenth of modern levels. around 0.8–0.6 Ga is indicated by increased levels of marine sulfate • The two rises in oxygen levels, at the beginning and end of the Proterozoic.

64) Protozoan Characteristics?

Protozoa are notable for their ability to move independently, a **characteristic** found in the majority of species. They usually lack the capability for photosynthesis, although the genus Euglena is renowned for motility as well as photosynthesis (and is therefore considered both an alga and a**protozoan**).

65) Note On Cambrian Explosion?

The Cambrian explosion generated a range of new body plans during a relatively short time interval.

66) Note On Cephalopodes?

A cephalopod is any member of the molluscan class Cephalopoda such as a squid, octopus or nautilus. These exclusively marine animals are characterized by bilateral body symmetry, a prominent head, and a set of arms or tentacles modified from the primitive molluscan foot.

67) Arthropods Features?

However, the members of this phylum, despite their incredible diversity and sheer numbers, share a number of important distinguishing characteristics.

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- Exoskeleton. ...
- Segmented Bodies. ...
- Jointed Appendages. ...
- Bilateral Symmetry. ...
- Open Circulatory System.

68) Among Polychaetes, Why Paddle Worms Have Complete Record?

Polychaetes: A diverse polychaete fauna has been described from the Burgess Shale • Even contains Canada spinosa • Similar to some living polychaetes. **Paddle worms:** most complete fossil record • Record is enhanced by the relatively common preservation of elements of the phosphatic jaw: scolecodonts.

69) Island Geography?

Islands are areas of land that are not connected to a continent and are surrounded by water. Small **islands** are sometimes called cays, keys, or islets. A group of**islands** is often called an archipelago.

There are two main type of **islands**; continental **islands** and oceanic **islands**.

70) Zones Of Fossil?

Index **fossils** (also known as guide **fossils** or indicator**fossils**) are **fossils** used to **define** and identify geologic periods (or faunal stages). Index **fossils** must have a short vertical range, wide geographic distribution and rapid evolutionary trends.

71) Bangiomorpha?

Bangiomorpha pubescens is a red alga. It is the first known sexually reproducing organism.

A multicellular fossil of **Bangiomorpha** pubescens was recovered from the Hunting Formation in Somerset

Island, Canada that strongly resembles the modern red alga Bangia despite occurring in rocks dating to 1,047 million years ago.

72) Nanoplankton?

Like other phytoplankton, **nanophytoplankton** are microscopic organisms that obtain energy through the process of photosynthesis and must therefore live in the upper sunlit layer of ocean or other bodies of water. ... The term**nanophytoplankton** is derived from the far more widely used term **nannoplankton/nanoplankton**.

73) Trilobitomorpha?

- Highly derived arthropods
- Lack specialized mouthparts
- Tagmata comprising the cephalon, thorax and pygidium
- Trilobitomorph appendages. Mainly the trilobites and over 15,000 species are known
- Common: throughout the Paleozoic
- Extinction: end of the Permian.

74) Island Biogeography?

Insular **biogeography** or **island biogeography** is a field within **biogeography** that examines the factors that affect the species richness of isolated natural communities. The theory was originally developed to explain species richness of actual **islands**, principally oceanic.

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75) **What Are Plesiomorphies?**

In phylogenetic, a **plesiomorphy**, symplesiomorphy or symplesiomorphic character is an ancestral character (trait state) shared by two or more taxa - but also with other taxa linked earlier in the clade (that is, having an earlier last common ancestor, with them, than theirs).

76) **Biostratigraphy?**

The branch of stratigraphy concerned with fossils and their use in dating rock formations.

77) **Biozones Range?**

Biostratigraphic unit or **biozones** are intervals of geological strata that are **defined** on the basis of their characteristic fossil taxa. ... The International Commission on Stratigraphy defines the following types of **Biozones**: **Range** zones, subdivided into taxon-**range**zones and concurrent-**range** zones.

78) **Triploblastic Body Plan?**

Few basic body plans have appeared in the fossil record • most animals have triploblastic architecture • Three fundamental body layers.

79) **Write the name of animal groups who hits by end permian mass extinction event?**

The end-Permian mass extinction was the largest of all time • Probably caused massive volcanic eruptions, leading to acid rain and global anoxia**. (ppts)**
During the period a number of animal groups became extinct, including the trilobites, tabulate and rugose corals, and blastoids (see mass extinction). **Amphibians** and **reptiles** continued to be the dominant land animals and gymnosperms replaced ferns, clubmosses, and horsetails as the dominant **plants**. "Permian. **(google)**

80) **What are the Trilobitomorpha? what are their different body parts?**

3 Highly derived arthropods
• lack specialized mouthparts
• Tagmata comprising the cephalon, thorax and pygidium
• Trilobitomorph appendages

• Common: throughout the Paleozoic
• Extinction: end of the Permian

81) **What is "Great Ordovician Biodiversification Event" (GOBE)?**

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE. Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels

82) **Important Definitions (02)**

Burgess Shale: The Burgess Shale is a fossil-bearing deposit exposed in the Canadian Rockies of British Columbia, Canada. It is famous for the exceptional preservation of the soft parts of its fossils. At 508 million years old, it is one of the earliest fossil beds containing soft-part imprints
Lagerstatten: A Lagerstätte (German: [ˈlaːɡɐˌʃtɛtə], from Lager 'storage, lair' Stätte 'place'; plural **Lagerstätten**) is a sedimentary deposit that exhibits extraordinary fossils with exceptional preservation—sometimes including preserved soft tissues.
Lithostratigraphy: Lithostratigraphy is a sub-discipline of stratigraphy, the geological science

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associated with the study of strata or rock layers. Major focuses include geochronology, comparative geology, and petrology. In general a stratum will be primarily igneous or sedimentary relating to how the rock was formed.

Signor-Lipps Effect: The **Signor–Lipps effect** is a paleontological principle proposed by Philip W. **Signor** and Jere H. **Lipps**which states that, since the fossil record of organisms is never complete, neither the first nor the last organism in a given taxon will be recorded as a fossil.

Chronostratigraphy: **Chronostratigraphy** is the branch of stratigraphy that studies the age of rock strata in relation to time. The ultimate aim of **chronostratigraphy** is to arrange the sequence of deposition and the time of deposition of all rocks within a geological region, and eventually, the entire geologic record of the Earth.

Tophonomy: Taphonomy is the study of how organisms decay and become fossilized.

Coenosis: An ecosystem, originally **defined** by Tansley (1935), is a biotic community (or biocenosis) along with its physical environment (or biotope). In ecological studies, biocenosis is the emphasis on relationships between species in an area.

Cladistics: Cladistics is an approach to biological classification in which organisms are categorized in groups based on the most recent common ancestor.

Fidelity: The **fidelity** of a DNA polymerase is the result of accurate replication of a desired template. ... High-**fidelity** PCR utilizes DNA polymerases that couple low misincorporation rates with proofreading activity to give faithful replication of the target DNA of interest.

83) What is great Ordovician biodiversity event (GOBE)?

Great Ordovician biodiversification event (GOBE), • Evolutionary radiation of animal life during the Ordovician Period, 40 my after the CE. Followed a series of Cambrian– Ordovician Extinction events • Fauna went on to dominate the Palaeozoic • Marine diversity increased Ordovician diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels

84) Niche with Example?

The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply help to define what ecologists call its **niche**. But the concept of the niche goes beyond the basic physics and chemistry of its habitat and covers all aspects of how the organism makes a living. It includes the food an animal requires, but also encompasses the way in which it acquires that food. The kestrel is a bird that hunts small mammals by day, whereas an owl performs a similar activity by night. Swallows catch aerial insects by day, and bats have the same feeding strategy at night

85) South America biogeography / History of South America? South America

The history of the biota of South America has been dominated by the effects of the great earth-engine of plate tectonics. Some of these effects were caused by the westward drift of the continent, which led to mountain building and consequent climatic changes, but most were the result of changes in its relationship with North America. As noted in the description of the origin of the Caribbean islands in the 'The West
Indies' section, there had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the Mid-Cretaceous, accompanied by some early types of marsupial and placental mammal. But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken.
This led to cycles of immigration, isolation and evolution, and final new immigration and extinction.**The**

Early Cenozoic:

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The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere. However, in the Late Cretaceous, South America had a mammal fauna o strange marsupials and of ungulate placentals quite unlike anything known elsewhere in the world. Alfred Wallace, one of the founders of zoogeography noted that the range boundaries of Amazonian birds seemed to coincide with the pattern of the rivers. It is fascinating that recent research shows that closely related taxa of some birds are to be found on either side of rivers throughout Amazonia.

86) **Geographical and Geological History of New Candolle / biogeography? New Caledonia**

Geological studies show that New Caledonia is the smallest (about 1600 km2) surviving fragment of Gondwana, having rifted away from Australia in the Late Cretaceous 80–65 mya and reaching its present position 1500 km east of it 50 mya. It is a hotspot for diversity: 77% of its flowering plant species are endemic to the island, as are 98% of the species of the flowering plant family Sapotaceae.

In view of the large amount of research that shows the frequency of long- distance dispersal in such scenarios, most biogeography's today believe that this was the mechanism of arrival of the ancestors of the **New Caledonian taxa.**

Geological studies show that the island was submerged for long periods in the **Paleocene–Eocene**, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date.

Panbiogeographers, however, **reject long- distance dispersal**. They therefore have to theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau.

Biogeographers in Sweden and France have investigated the history of the New Caledonian species of the angiosperm family Sapotaceae. They found that these resulted from nine separate colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia. New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

87) **Kleptoarasitisim with example?**

Kleptoparasitism (literally, parasitism by theft) is a form of feeding in which one animal takes prey or other food from another that has caught, collected, or otherwise prepared the food, including stored food (as in the case of cuckoo bees, which lay their eggs on the pollen masses made by other bees; food resources could also be in the form of hosts of parasitic or parasitoid wasps). The term is also used to describe the stealing of nest material or other inanimate objects from one animal by another.

One very widespread duck in both continents is the gadwall. It feeds mainly on submerged vegetation, which is most abundant in nutrient- rich (eutrophic) waters. It cannot dive, so it feeds by upending, and this generally restricts it to the shallow parts of a lake, where it may encounter some competition from other water birds, such as the mallard. Coots (both Eurasian coot and the American coot dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called **kleptoparasitism**, and it is an effective way of widening the niche of the gadwall.

88) **Modern bird evolved how and when / when modern birds appear?**

Modern groups of birds appeared in the latest Cretaceous and Early Tertiary, including flightless ratites ancestors of water birds, penguins and birds of prey. The perching birds or songbirds, consisting of 5000 species today, radiated in the Miocene

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89) History of old world mammal in fauna Africa?

Africa contains the best known and most distinctive of the Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous. This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a super order, Afrotheria, that is endemic to Africa. The early horse *Hipparion*, which had evolved in North America, appeared in both Eurasia and Africa, while rhinoceros, hyenas and sabre- toothed cats dispersed from Africa to Eurasia. The elephants originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene. The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

90) Biotic replacement?

Biotic Replacements are an obvious feature of the history of life. These are times when one group of plants or animals replaces another. The replacement of brachiopods by bivalves is a famous example. This had always been seen as a progressive process: the common view is that brachiopods are less adaptable than bivalves, and they clearly succumbed to long-term competition, perhaps lasting for tens or hundreds of millions of years.

Thank You !

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